MD SHAHIDUL KHAN PAPPO

Trustworthy AI for Healthcare | Vision-Language Models | LLM Agents

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LinkedIn GitHub Google Scholar ResearchGate

PROFESSIONAL SUMMARY

Computational AI Researcher and Computer Science graduate specializing in trustworthy, explainable, and reliable medical imaging and biomedical AI applications. Proven research track record with 4+ peer-reviewed publications, including Q_1 indexed journal articles in **IEEE Access** and **Artificial Intelligence and Applications**. Expertise spans calibrated federated learning under distribution shifts, post-hoc interpretability (XAI), LLM and AI-agent systems, and production MLOps deployment.

EDUCATION

B.Sc. **American International University-Bangladesh (AIUB)** Jun 2021 – Apr 2026

B.Sc. in Computer Science and Engineering (CSE) | CGPA: 3.05/4.00

– **Major** in Computational Theory | **Graduated:** April 2026

– **Relevant Coursework:** Artificial Intelligence, Machine Learning, Statistics, Linear Algebra, Computer Vision & Pattern Recognition, Natural Language Processing (NLP), Human–computer interaction (HCI)

UNDERGRADUATE THESIS

Title: **Robust Ensemble Learning for Multi-Class Cognitive State Classification of EEG Data**

– **Thesis Supervisor:** Prof. Dr. Md. Asraf Ali

– **Note:** Research manuscript currently under preparation for peer-reviewed journal submission.

RESEARCH INTERESTS

• Multimodal learning • Machine Learning • Medical imaging • Trustworthy AI • VLM

PROFESSIONAL & RESEARCH EXPERIENCE

Assistant Investigator Nov 2026 – Present

Artificial Intelligence Research & Innovation Lab (AIRIL), Dhaka

Focus Conduct research in Healthcare AI, Machine Learning, Computer Vision, LLMs, and VLMs. Develop and evaluate deep learning models and LLM/VLM-based systems. Contribute to journal and conference publications and deploy AI applications using Flask, Django, and Streamlit.

– **Leadership:** Lead research teams as **Research Lead, McCarthy Lab**.

– **Mentoring:** Mentor junior students in programming, machine learning, and research.

Research Assistant (RA) Dec 2025 – Oct 2026

Artificial Intelligence Research & Innovation Lab (AIRIL), Dhaka

Focus Conducted Healthcare AI and Computer Vision research, developed and evaluated deep learning models, contributed to publications, and supported AI application development.

Junior Research Assistant (Internship) Aug 2025 – Dec 2025

Artificial Intelligence Research & Innovation Lab (AIRIL), Dhaka

Focus Supported healthcare AI research through data preprocessing, literature reviews, experimentation, and documentation.

RESEARCH PUBLICATIONS

Journal Article

2026 **A Novel Explainable Deep Learning Model for Early-Stage Brain Tumor Classification: Multi-Level Feature Fusion from Merged MRI Datasets.** *Artificial Intelligence and Applications (AIA)*, 2026. **Q_1 Journal** DOI: 10.47852/bonviewAIA62026182

2025 **NefroAI:** An Explainable and Real-Time Framework for Predicting CKD Using Diverse ML Models and Different Feature Selection Techniques. *IEEE Access*, Dec 2025. **Q_1 Journal** DOI: 10.1109/ACCESS.2025.3649006

Conference Papers

2026 **MindBeaconAI:** An Explainable Machine Learning Framework for Student Stress Level Prediction and Real-Time Deployment. *2026 IEEE 3rd COMPAS Conference*, 2026. **Accepted for publication.**

2025 **OvaSignal AI:** An Explainable and Real-Time Machine Learning Framework for Early Prediction of Ovarian Cancer. 11th *WIECON-ECE*, 2025. **DOI:** 10.1109/WIECON-ECE69386.2025.11526637

2025 **RenalAI:** An Explainable Machine Learning Framework for Real-Time Early Prediction of CKD Using Electronic Health Records. 28th *ICCIT*, 2025. **DOI:** 10.1109/ICCIT68739.2025.11490343

Under review

2026 **PulmoSense AI:** An Explainable Multi-Model Approach for Lung Cancer Prediction in Smart Healthcare Systems. *Health Informatics Journal*, 2026. **Submitted June 10, 2026; Under Peer Review.**

2026 **An Agentic Vision-Language-Enabled DEGT-Net for Retinal Vessel Segmentation with Federated Calibration and Multi-Method Explainability.** *Artificial Intelligence in Medicine*, 2026. **Submitted August 21, 2026; Under Peer Review.**

CONFERENCE PARTICIPATION

2025 Presented, 28th ICCIT 2025, IEEE Bangladesh Section, Long Beach Hotel, Cox's Bazar, Bangladesh, December 19–21, 2025.

2025 Presented, 11th WIECON-ECE 2025, IEEE Bangladesh. Long Beach Hotel, Cox's Bazar, Bangladesh, December 22–23, 2025.

2026 Presented, 3rd IEEE COMPAS 2026, IEEE Computer Society Bangladesh Chapter. University of Dhaka, Dhaka, Bangladesh, October 9–10, 2026.

ACADEMIC SERVICE

2026 **Reviewer**, 10th International Conference on Computer Science and Application Engineering (CSAE 2026), October 17–19, 2026, Shanghai, China.

TECHNICAL PROJECTS

Projects Information

2026 **RetinaAgent: Multimodal Healthcare AI System** (GitHub) – Multimodal retinal image analysis system integrating Attention U-Net, Explainable AI (XAI), Vision-Language Models (VLMs), and LangChain-based agents with RAG for interactive, explainable analysis.

2026 **EchoMind: AI-Powered Agentic Assistant** (GitHub) – LLM application with RAG, memory, document analysis, and tool integration using LangChain, FAISS, HuggingFace embeddings, and Google Gemini.

2026 **Pathok: RAG Document Intelligence System** (GitHub) – Retrieval-Augmented Generation (RAG) application using LangChain, FAISS, HuggingFace embeddings, and Groq LLM for document-based question answering with persistent vector search.

2025 **ModelLensAI** (GitHub) – Production-ready MLOps platform supporting model deployment, benchmarking, inference, latency analysis, experiment tracking, and REST API serving using Flask, Docker, MLflow, PyTorch, TensorFlow and ONNX.

TECHNICAL SKILLS

Track 1 **Programming** – Python | C/C++ | C# | SQL | MySQL | MongoDB.

Track 2 **ML & Deep Learning** – PyTorch | TensorFlow | Keras | Scikit-learn | Transformers | Fine-Tuning | Statistics | Feature Engineering.

Track 3 **Computer Vision** – Medical Image Analysis | Image Segmentation | OpenCV | YOLO | Roboflow.

Track 4 **LLMs & VLMs** – LangChain | LangGraph | RAG | Hugging Face | Prompt Engineering | Fine-Tuning.

Track 5 **Explainable AI (XAI)** – SHAP | LIME | Grad-CAM | Model Interpretability.

Track 6 **Backend Development** – FastAPI | Flask | Django | REST APIs.

Track 7 **Data Analysis & MLOps** – NumPy | Pandas | Power BI | Docker | MLflow | Kubernetes | CI/CD.

Track 8 **Research Tools** – Jupyter Notebook | Google Colab | Kaggle | Git/GitHub | LaTeX.

ACHIEVEMENTS

Awards and Honors

2025 **Finalist**, International Computer Science Competition (ICSC), Nov 2025 — Ranked in the top **10%** of all global participants; recognized for strong problem-solving, algorithmic thinking, and innovation in the Senior Category. **Verify-ID:** F-2025-FF864199CA5

2025 **Qualifier**, International Youth Math Challenge (IYMC), Dec 2025 — Successfully completed the Qualification Round. **Verify-ID:** PR-2025-100C48C27D23

LEADERSHIP & ACTIVITIES

i. Firefox at AIUB — Club Lead Peer — *Oct 2023 – Dec 2024, Dhaka, Bangladesh*

ii. BASIS Students' Forum — Executive Member — *Nov 2022 – Oct 2023, Dhaka, Bangladesh*

iii. AIUB Computer Club — General Member / Video Editor — *Apr 2022 – Dec 2024, Dhaka, Bangladesh*

VOLUNTEERING

AIUB Computer Club — **Volunteer** — AIUB Cyber Gaming Fest 2022, Dhaka, Bangladesh

BASIS Students' Forum (AIUB Chapter) — **Volunteer** — BASIS SoftExpo 2023, Purbachal, Dhaka, Bangladesh

LANGUAGES

Bengali (Native) | English (Professional Working Proficiency)